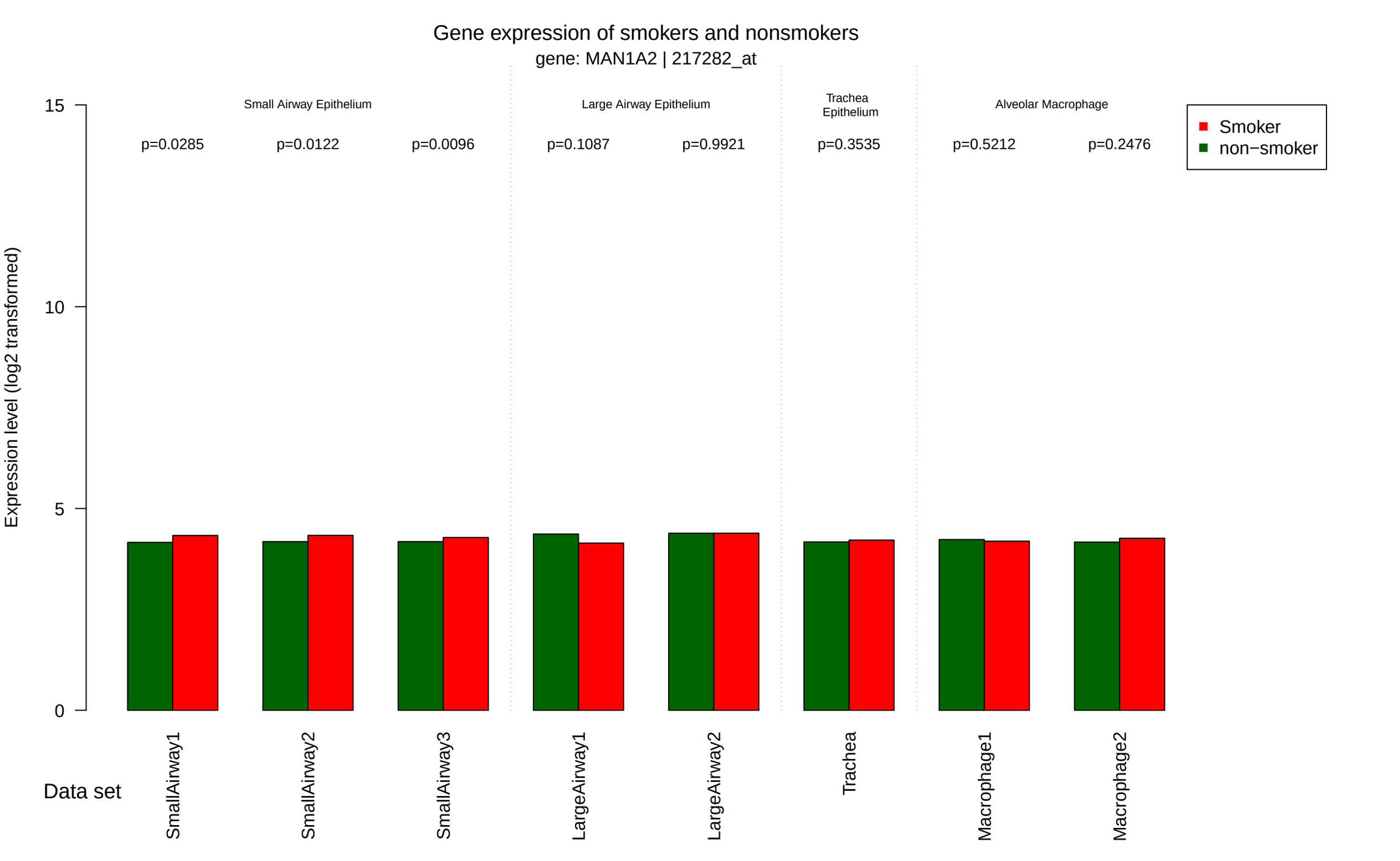
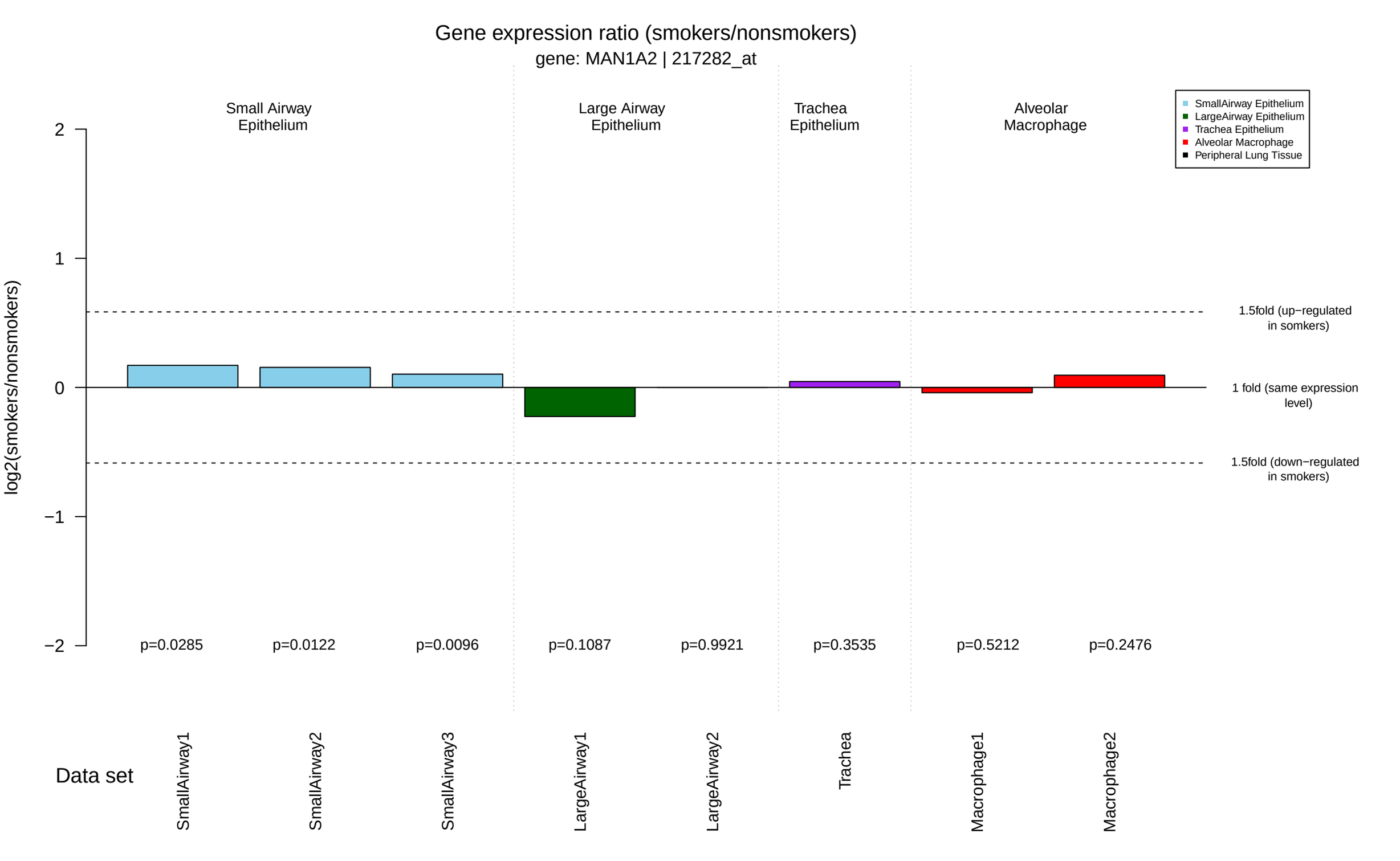
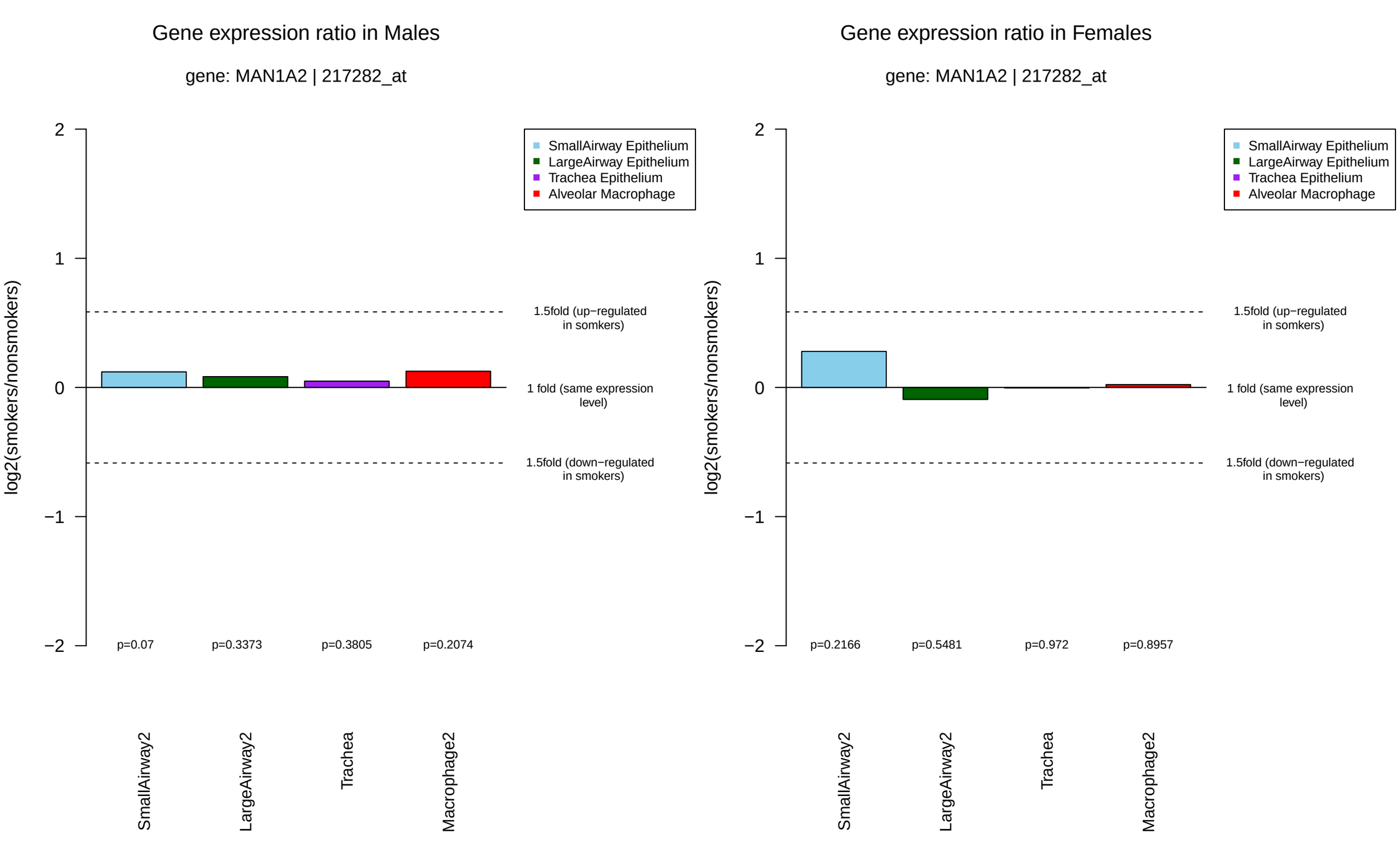
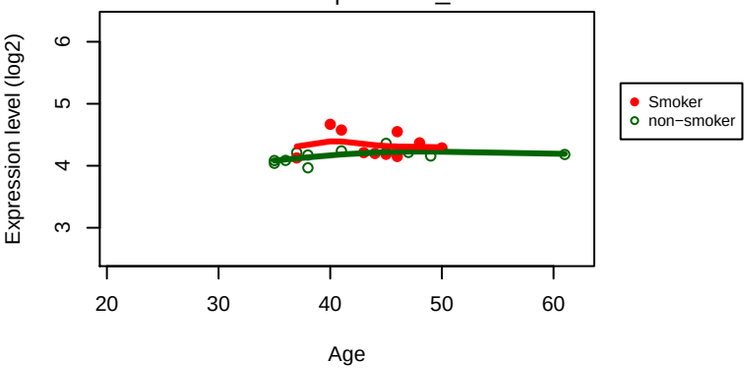
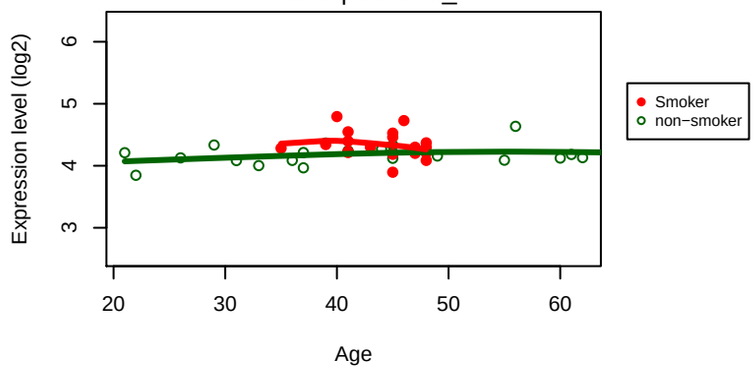
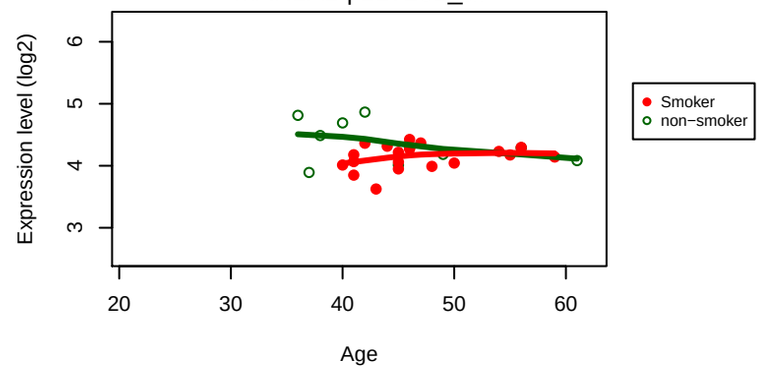

Gene expression of smokers and nonsmokers
gene: MAN1A2 | 217282_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0285
p=0.0122
p=0.0096
p=0.1087
p=0.9921
p=0.3535
p=0.5212
p=0.2476
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

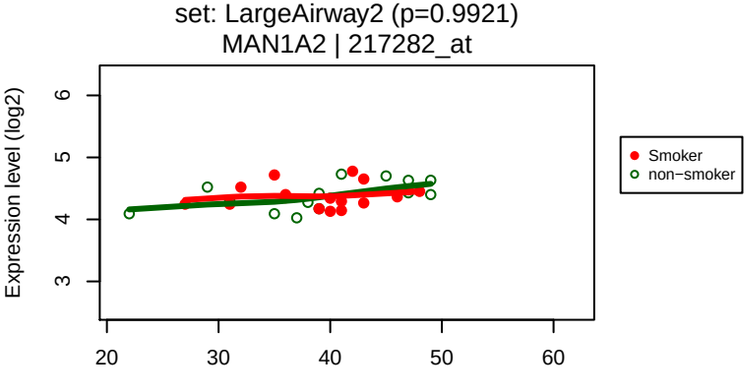

Gene expression ratio (smokers/nonsmokers)
gene: MAN1A2 | 217282_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0285
p=0.0122
p=0.0096
p=0.1087
p=0.9921
p=0.3535
p=0.5212
p=0.2476
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

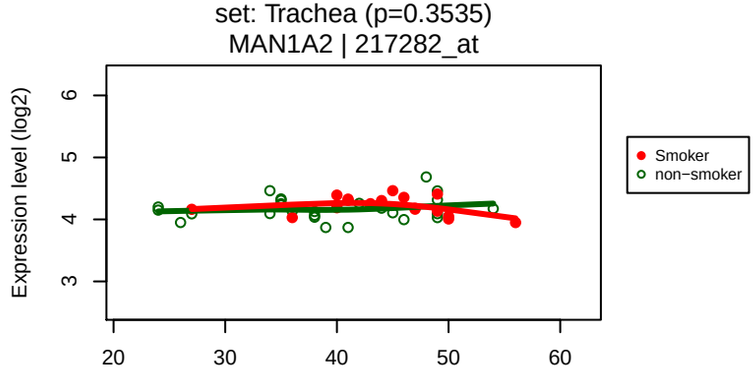

Gene expression ratio in Males
gene: MAN1A2 | 217282_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.07
p=0.3373
p=0.3805
p=0.2074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAN1A2 | 217282_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2166
p=0.5481
p=0.972
p=0.8957
SmallAirway2
LargeAirway2
Trachea
Macrophage2

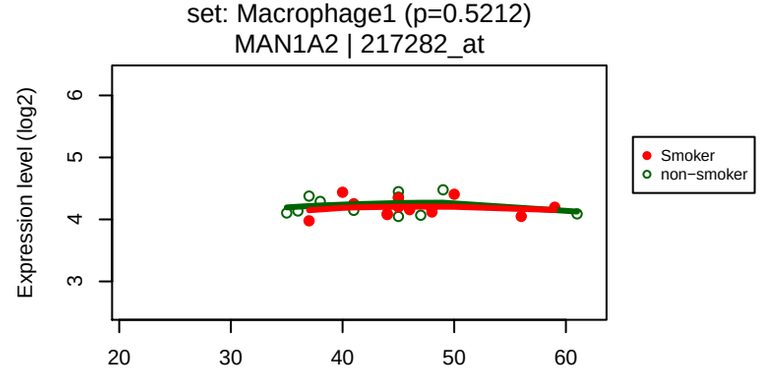

set: SmallAirway1 (p=0.0285)
MAN1A2 | 217282_at
Expression level (log2)
Age
Smoker
non−smoker

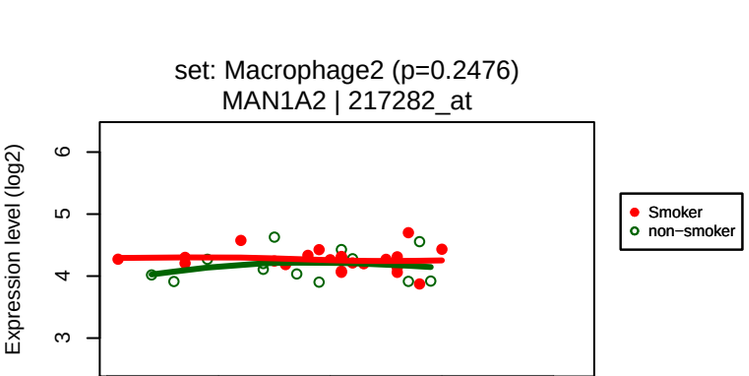

set: SmallAirway2 (p=0.0122)
MAN1A2 | 217282_at
Expression level (log2)
Age
Smoker
non−smoker

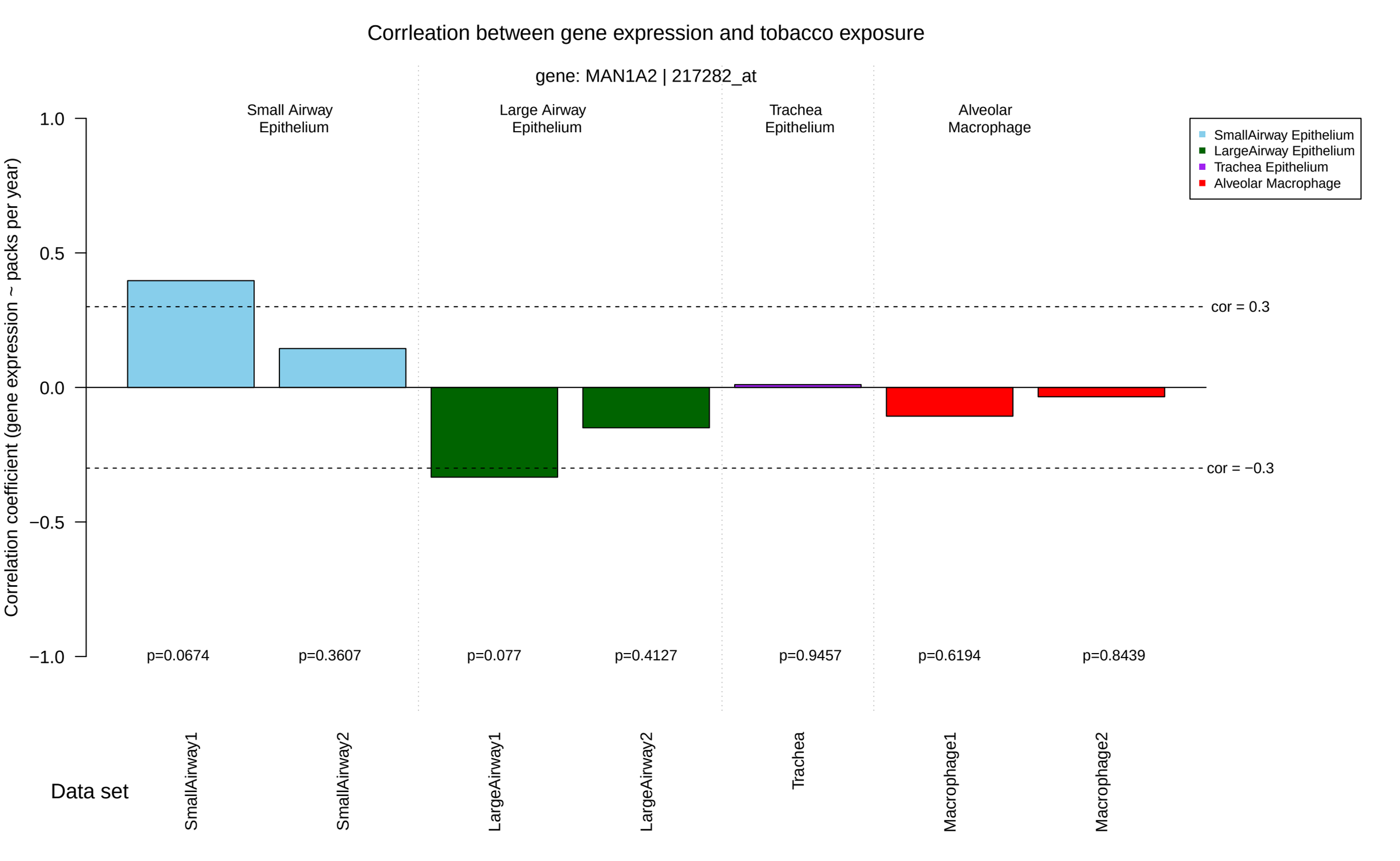

set: LargeAirway1 (p=0.1087)
MAN1A2 | 217282_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9921)
MAN1A2 | 217282_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3535)
MAN1A2 | 217282_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5212)
MAN1A2 | 217282_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2476)
MAN1A2 | 217282_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAN1A2 | 217282_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0674
p=0.3607
p=0.077
p=0.4127
p=0.9457
p=0.6194
p=0.8439
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2